

# Smart Home Energy Efficiency Handbook

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## Overview

Smart homes use intelligent automation systems to reduce electricity consumption and improve operational efficiency.

## Efficient Appliances

Energy-efficient appliances consume less electricity and reduce long-term utility bills.

## Cooling Optimization

Air conditioners should ideally operate between 24°C and 26°C for balanced comfort and efficiency.

## Lighting Systems

LED lighting significantly reduces power consumption compared to traditional incandescent bulbs.

## Automation

Motion sensors, smart thermostats, and scheduling systems help automate household energy management.

| Metric                       | Example Value |
|------------------------------|---------------|
| Average Solar Efficiency     | 21%           |
| Recommended AC Temperature   | 24°C          |
| Peak Solar Hours             | 10 AM - 3 PM  |
| Average EV Charging Duration | 4-6 Hours     |